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FFT Implementation on FPGA for 5G Networks

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ABSTRACT

The main goal of this thesis will be the design and implementation of a 2048-point FFT on an FPGA through the use of VHDL code.

The FFT will use a butterfly Radix-2 architecture with focus on the comparison of the parameters between the system with different Wordlengths, Coefficient Wordlengths and Symbol Error rates as well as different modulation types, comparing 64QAM and 256QAM for the 5G system.

This implementation will replace an FFT function block in a Matlab based open source 5G NR simulator based on the 3GPP 15 standard and simulate spectrum, MSE payload, and SER performance.

Keywords: FFT, OFDMA, Physical design, FPGA, 5G, VHDL, CP-OFDMA, Radix-2, 2048-Point.

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NOMENCLATURE

5G Related Nomenclature

<i>OFDMA</i>	Orthogonal frequency-division multiple access
<i>OFDM</i>	Orthogonal frequency-division multiplexing
<i>CP</i>	Cyclic prefix
<i>QAM</i>	Quadrature amplitude modulation
<i>LTE</i>	Long-Term Evolution
<i>NR</i>	New Radio
<i>3GPP</i>	3rd Generation Partnership Project
<i>SNR</i>	Signal-to-noise ratio
<i>PAPR</i>	Peak-Average-Power-Ratio
<i>UE</i>	User Equipment
<i>SER</i>	Symbol Error Rate
<i>CFO</i>	Carrier Frequency Offset
<i>MSE</i>	Mean Square Error

FFT Related Nomenclature

<i>FFT</i>	Fast Fourier Transform
<i>IFFT</i>	Inverse Fast Fourier Transform
<i>DFT</i>	Discrete Fourier Transform
<i>VHDL</i>	(Very High Speed Integrated Circuit) Hardware Description Language
<i>FPGA</i>	Field-Programmable Gate Array
<i>WL</i>	Data Wordlength
<i>CWL</i>	Coefficient Wordlength

Other symbols

<i>IoT</i>	Internet of Things
<i>mMTC</i>	Massive Machine-type Communications

CHAPTER 1

INTRODUCTION

1.1 Motivation

With the increase of devices in the household and industry applications, a large amount of strain to the networks is caused. As such engineers always look for new methods to increase the accessibility and speed of such networks.

The main response to this increased need for accessibility and the limitations of the former standards is the new fifth-generation cellular network technology (5G).

As such the 5G network needs to be able to provide these services as fast as possible, which leads to the need to do different studies on ways to increase the speed and efficiency of such systems.

This thesis proposes to achieve this exact aspect, with a focus on the FFT block. The choice was made due to the FFT being one of the most important blocks in most digital communication systems, and as such it is an automatic candidate for improvement in design and research as well as different implementation routes.

This leads us to the main motivation of this project, the implementation of the specified FFT block for this new communication system with a focus on wordlength reduction through optimizations in VHDL.

1.2 Aim

The principal aim of this thesis is the use of a 5G NR simulator[1] with a VHDL designed FFT block to describe the behavior of the system, with focus on the following parameters: data wordlength, coefficient wordlength, modulation type, and SER.

The results are afterwards compared to the baseline simulations made when using the 5G simulator[1] in a normal test case environment.

1.3 Research Questions

1. What limitations need to be applied to the FFT VHDL block for its correct functionality while being used together with the 5G NR simulator?
2. How does the FFT function which was implemented in Matlab programming compare when it is replaced with the designed VHDL FFT block?
3. What values can be achieved with this FFT block when focusing on the following parameters: data wordlength, coefficient wordlength, modulation type, and SER.

1.4 Approach

To answer the stated research questions the following steps will be taken:

Chapters 2 and 3 will introduce the theoretical aspects of both systems that form the foundation of this thesis (5G and FFT block) and describe the parameters which need to be discussed.

The second step will follow with the description of the implementation of the FFT system as well as characteristics which relate to the correct functionality of the 5G NR system together with focus on the FFT block.

The Simulation chapter will describe the benchmark used for the implemented FFT block to assure its correct functionality and test the system efficiency of the system when comparing to the new approach to the simulator.

Furthermore the last two chapters will discuss the substitution of the FFT block and the results achieved while comparing them to the simulations from the original Matlab

system.

This will lead to the discussion of the research questions which were originally stated, while also providing a look at possible future work which can be done in this area.

1.5 Delimitations

Due to this project using an open-source simulator provided by a 3rd party, this automatically creates constraints to the desired system while trying to combine the software with the FFT.

Examples of constraints contained by the system relate to channel simulation which is not given by the simulator and the simulation only considering a small number of users, which would be different from a real-life implementation.

The most important delimitation to the 5G system is the noise caused by the AWGN channel as well as other points that create noise. This allows the system to give an SER of 0, for the simulations so that the only factor affecting the simulations done in this project are the changes done to the FFT implementation and its parameters. Another compromise that had to be made is the use of only one packet during the simulation compared to a real-life test case where multiple packets need to be sent.

Description of the used standard in connection to the Fourier transform block

2.1 5G NR Standard

In the last years, the exponential growth of connected devices and traffic volume has created a need to expand technologies in the area of digital communication to allow the increasing number of types of equipment to function effectively together.

The International Telecommunication Union describes the technical requirements which need to be achieved until the year 2020 in the IMT-2020 Standard for communication systems, and as a response to this need, the 5G standard is being developed.[2]

The 5G NR Standard was released in 2018 representing version 3GPP 15, with the upcoming version 16 being prepared for release in late 2019 or the beginning of 2020. This version is backwards compatible as well with the older 2G, 3G, and 4G technologies and has a higher spectrum range which starts at 1GHz and goes up to the mm-Wave range until 52.6 GHz, whereas to the older standard was working only at frequencies lower than 6 GHz.[3]

2.2 Physical Layer

In the design of this thesis, the physical layer is the key element of concern. The main characteristics of the physical layer deal with modulation and demodulation, mapping of signals in time and frequency, and usage of multiple antennas.

5G NR supports Quadrature Amplitude Modulation up to 256 points as the LTE standard and in the future may also include 1024-QAM, but with the standard version currently available, it is not available.[3]

Table (2.1): Scalable OFDM numerology for 5G NR (3GPP Release 15)[3]

Subcarrier Spacing	15 kHz	30 kHz	60 kHz	120 kHz
Frequency Band	0.45-6GHz	0.45-6GHz	0.45-6GHz, 24-52.6GHz	24-52.6GHz
OFDM symbol duration	66.67us	33.33us	16.67us	8.33us
Cyclic prefix duration	4.69us	2.34us	1.17us	0.59us
OFDM symbol with CP	71.35us	35.68us	17.84us	8.91us
Maximum bandwidth	50 MHz	100 MHz	200 MHz	400 MHz

Table 2.1 enumerates the relation between subcarrier spacing and the OFDM characteristics which describe the implementation of the system. It should be taken into account that the subcarrier spacing of 60 kHz is the only one that can describe both 0.45-6 GHz and 24-52.6 GHz intervals.

Beside the above presented parameters another important value is the maximum number of active subcarriers. The maximum number of subcarriers is achieved at the maximum bandwidth of 400 MHz for an FFT value of 4096, where the number of used active subcarriers is 3276. Although 3276 is the maximum currently achieved number, the largest possible value is 3300 subcarriers.[4]

2.3 Waveform

The used waveform type for 5G NR is CP-OFDM on both downlink and uplink. This choice creates a reduced complexity model for the communication system between the central system and the local networks.

The number of active subcarriers is 3276 for all numerologies, which leads to the maximum bandwidth enabled by the parameters described in Table 2.1.[5]

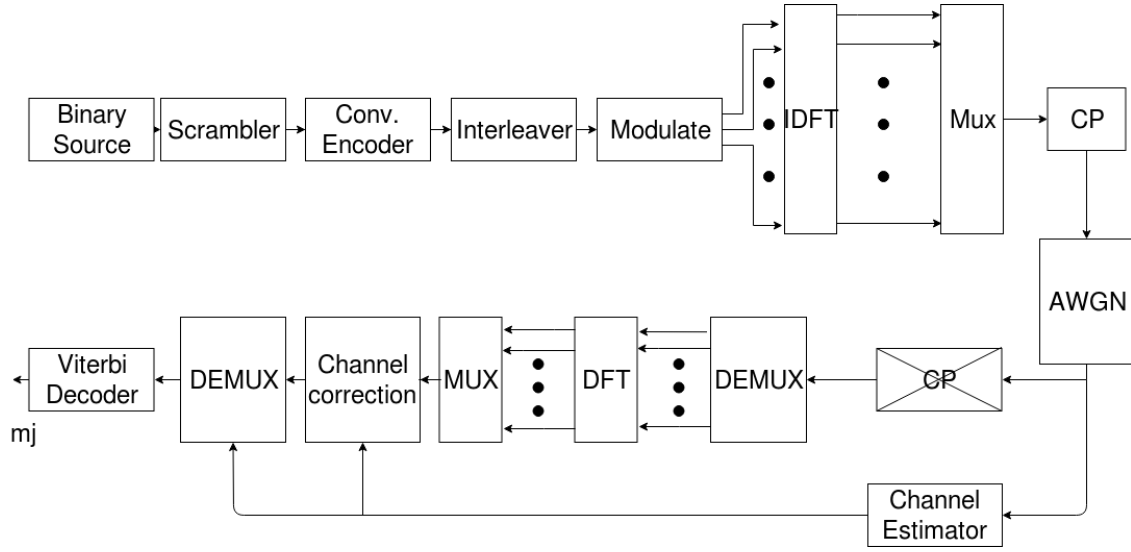


Figure (2.1): Block diagram of a OFDM Transceiver

Figure 2.1 is described as a reference point in comparing the block diagram that will be used in the simulation of the system to a generic OFDM Transceiver for the 802.11a system.[6]

An even more complex design of the transceiver for 4G is presented in [7] but for the requirements of this thesis, the 802.11a is a better model of comparison.

This is done as a consequence of the fact that several blocks have been simplified or included in other blocks in the 5G simulator, and as such does not directly respect the standards block view expected from the system.

The multi-user scenario where OFDMA Figure 2.2 is used will also be discussed as it significantly increases the performance of the system and reduces the wait time for users especially those which require a small amount of information. This case happens when the frequency is divided to give multiple users access to data during a period T.

This multi-user system follows the next pattern:

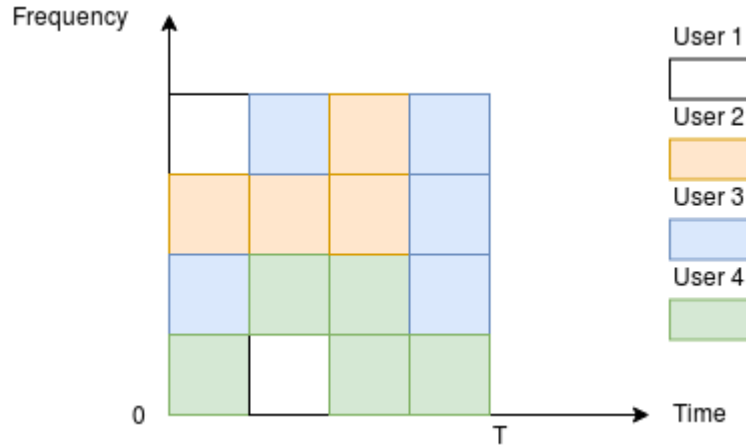


Figure (2.2): Possible distribution of frequency for a number of 4 users

The OFDM can be implemented using FFT/IFFT at the receiver/transmitter which is the main design block, where the used symbols(QAM) are mapped to the subcarriers by transforming the signal from serial to parallel, followed by an IFFT of size N, and back to a parallel to serial converter.

As such the FFT is a crucial element of these systems, where the standard's length in 3GPP Release 15 is between 64 and 4096 points.[4]

When discussing communication systems the parameters that are used by them are listed in Table 2.2:

Table (2.2): System Parameters for WLAN, 4G and 5G [8]

Standard	Waveform	FFT Sizes	Modulation	Maximum Bandwidth
802.11n/ac	OFDM	64-512	up to 256 QAM	160 MHz(512 point)
4G LTE/LTE-A	OFDM	128-2048/1536 12-2400	up to 256 QAM	30.72 MHz (2048 Point)
5G	OFDM	up to 4096	up to 256 QAM	400 MHz (4096 Point)

In such a system which contains a high modulation complexity, it must also contain a high QAM which leads to the requirement of precision which changes its range dynamically.

The system parameter description in Table 2.2 shows a trend of an increasing number of FFT transformation points over the years which continues to increase with each version of the standards.

This leads to the necessity of a proper implementation of the FFT processor especially related to the accuracy of the wordlengths of the system as the more points/closer the data is the higher the probability of error becomes, which leads to the discussion of this block in the next chapter.

CHAPTER 3

FAST FOURIER TRANSFORM THEORY

This Chapter will describe the theoretical aspects of the Fast Fourier Transform

3.1 Introduction

In the digital signal processing domain, the Fast Fourier transform represents one of the most important algorithms as it has a large variety of uses in different implementations, examples of these being: image processing, quantum computing or telecommunications. The Fast Fourier Transform represents a more efficient way of calculating the discrete Fourier transform.

The Discrete Fourier Transform changes the time plane of a system to the frequency plane, different signals being described by different frequency values.

It is defined by the following parameters with a number of equally spaced samples N :

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{\frac{(-j2\pi nk)}{N}} \quad (3.1.1)$$

While its inverse described by:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{\frac{j2\pi nk}{N}}, \text{ where } n = 0, 1, 2, \dots, N - 1 \quad (3.1.2)$$

The transform is periodic in ωT with a period of $\frac{2\pi k}{N}$, where the variable ωT is independent.

While its Twiddle factor is defined by:

$$W_N = e^{-j\frac{2\pi}{N}} \quad (3.1.3)$$

This Twiddle factor represents trigonometric coefficients that are multiplied by the data which are represented by the twiddle factor multiplication N , and the sample points of the DFT.

The DFT is executed in iterations with each component being computed separately, and as such, is not efficient.

The Fast Fourier Transform is a response to this problem, as it is a more efficient implementation of the DFT, which reduces the number of computations required. Instead of calculating each component separately, the transform splits the data into stages to be executed and combined at the end to achieve the result of the transform.

There are 2 classes of FFT algorithms:

1. Decimation in time (DIT) algorithm
2. Decimation in frequency (DIF) algorithm

The most common implementation of the FFT uses a butterfly architecture which is described by the figure 3.1.

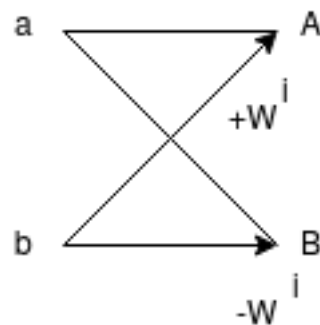


Figure (3.1): *Butterfly for decimation-in time*

Decimation in frequency is used in the implementation of the system with twiddle multiplication being applied to the second input before the butterfly stage, which reduces the number of multipliers needed.

Expanding the algorithm for multiple stages and inputs leads to:

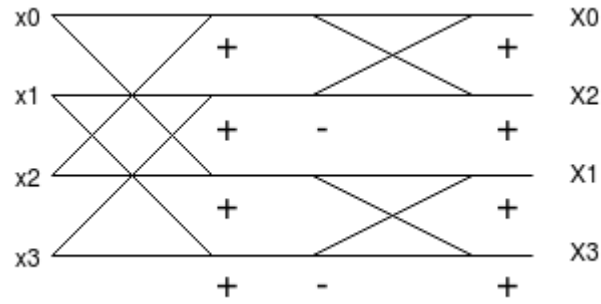


Figure (3.2): *Flow graph of the algorithm*

Chapter 2 specified that the maximum number of possible active subcarriers is of 3276, which can be implemented using an FFT of 4096 points, total memory of size 4096, and for each FFT point one complex adder, 2 complex multipliers and 1 complex subtractor.

The complexity of an N-point FFT operation is defined by:[4]

$$O(N \log N) \quad (3.1.4)$$

There are a large number of alternate implementations for the FFT, the main difference between them being the size of the processing elements related to butterflies and rotators.

Examples of algorithms for the implementation of FFT are:[3]

1. Cooley-Tukey FFT algorithm
2. Good Thomas FFT algorithm
3. Bruun's FFT algorithm
4. Rader's FFT algorithm

3.2 Cooley-Tukey FFT algorithm

The Cooley-Tukey FFT algorithm, represented in decimation-in-frequency, name defined by its usage of a Divide and Conquer approach in time domain where the N-point FFT is divided into stages: 2 stages in the first level, $4 \frac{N}{4}$ -point FFTs afterwards and so on.

This is the most common fast Fourier Transform algorithm used, the Cooley-Tukey algorithm requiring only $O(N \log_2(N))$ operations.

For very large N the difference in execution time is also very large between the new algorithm and the computation time of the DFT algorithm.

As an example, the time that is required for all complex multiplications in a 4096-FFT where it was assumed the one complex multiplication is equal to 4 real multiplications, and that a real multiplication takes 100ns leads to the following equation:

$$T_{mult} = 0.5 * N * \log_2(N) * 2 * 100ns = 0.5 * 4096 * \log_2(4096) * 2 * 100ns = 9.830 \text{ ms} \quad (3.2.1)$$

Which will result in a sample frequency of approximately 100kHz.

The inverse of the transform (IFFT) can be computed by switching the input sequence and calculating the FFT of the following:[4]

$$X'(0) = \frac{X(0)}{N} \quad (3.2.2)$$

$$X'(n) = \frac{X(N-n)}{N} \text{ for } n = 1, 2, \dots, N-1 \quad (3.2.3)$$

Address 0 stores the first value while the rest of the values have their read order changed(reverse bit), changing the order in which the bits are read from left to right which is run using standard signal processing with a complex sequence of length N .[4]

Calculation of the IFFT can also be implemented by changing the real and imaginary parts of the system at the input, applying the FFT block to it and changing back the outputs real and imaginary parts greatly reducing the amount of work.

3.3 Carrier

The FFTs implementation is directly linked to the 5G NR system and as such the study of spectrum flexibility was discussed earlier in Chapter 2 and the used frequency will be mentioned later, as it is needed in the implementation of the system.

Noteworthy are the parameters of Table 2.1 combined with the FFT size of 2048 which is of concern to the implementation of the system and as such the other values with different

subcarrier spacings than 120kHz will not be discussed further in the implementation chapter.

CHAPTER 4

METHOD

This chapter will deal with the practical aspect of the FFT implementation related to the specification of the 5G NR Standard

4.1 FFT Implementation

The main factor in the decision to implement an FFT algorithm in VHDL for FPGAs is caused by a large number of institutions and companies being interested in the implementation of this algorithm especially in wireless communication. Another factor is the fact that most implementations are done on Application-specific integrated circuits but not on FPGAs.

The implementation which is used the most is the Radix-2 butterfly, although of course there are examples of other cases of Radix-4 [9] or Radix-8,16 [5].

As stated previously this thesis will deal with the study of the FFT using Cooley-Tukey algorithm and Radix-2 butterfly architecture.

The proposed architecture of the 2048-FFT requires 11 Radix-2 cascaded stages, which is deduced from the following expression:

$$Number_of_stages = \log_2 N \quad (4.1.1)$$

This is combined with the usage of a pipeline architecture with the following top view:

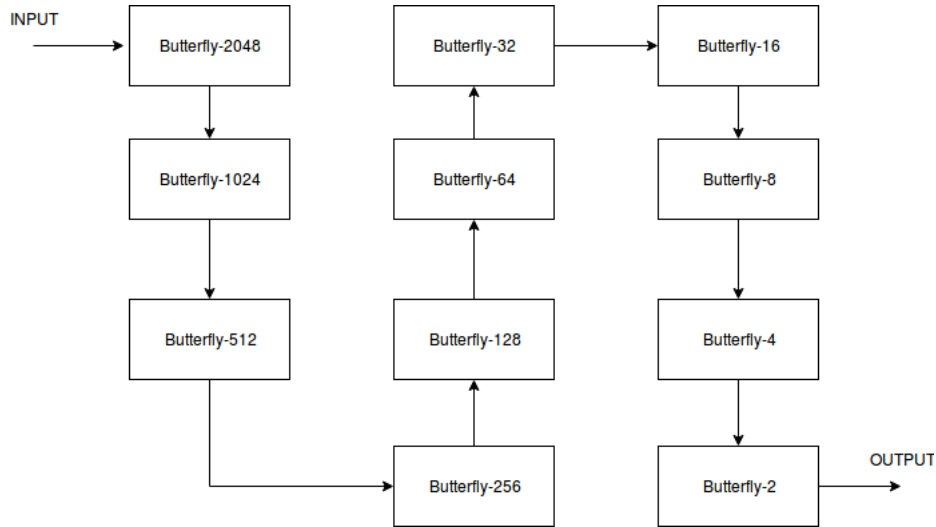


Figure (4.1): 2048-FFT Butterfly Division

The inputs of the FFT will use 2 fixed-point signed vectors of varying wordlengths between 6 and 24, where one vector describes the real part and the other the imaginary part of a number. Important is the fact that the initial integer part of the wordlength is always 4 and thus the fractional part is the one which is alternated when discussing wordlength.

The choice of using a integer wordlength of 4 was made after doing initial simulations on the 5G system and inspecting the values which are being input. Since none of the values were larger than 2 the wordlength of 4 was chosen for the fixed integer part.

For instance if the wordlength is 6 the fractional wordlength is of a length of 2, which means the maximum fractional wordlength is 20 in this system. To assure there are no errors in the integer part during addition between stages, the integer wordlength is always increased by 1 for each stage of the FFT.

Since VHDL does not provide an implementation of sin/cos needed to calculate the twiddle Factor, its results will be stored directly in a vector in the code and taken from there with a change of wordlength before the values are taken. The twiddle factors accuracy is represented by the coefficient wordlength.

The input values are read as "real" types for both the real and imaginary parts of the complex number, which are generated by the 5G system. These values are then truncated since the wordlength of Matlab is larger than the ones proposed in the measurements.

The FFT systems top level view could be described as follows for a decided integer wordlength resulted from the simulations of 4 bits for the real and imaginary part, and an

integer wordlength of 2 bits for the coefficient wordlength. For the fractionary wordlength M and N will be given alternating values between 4 and 24.

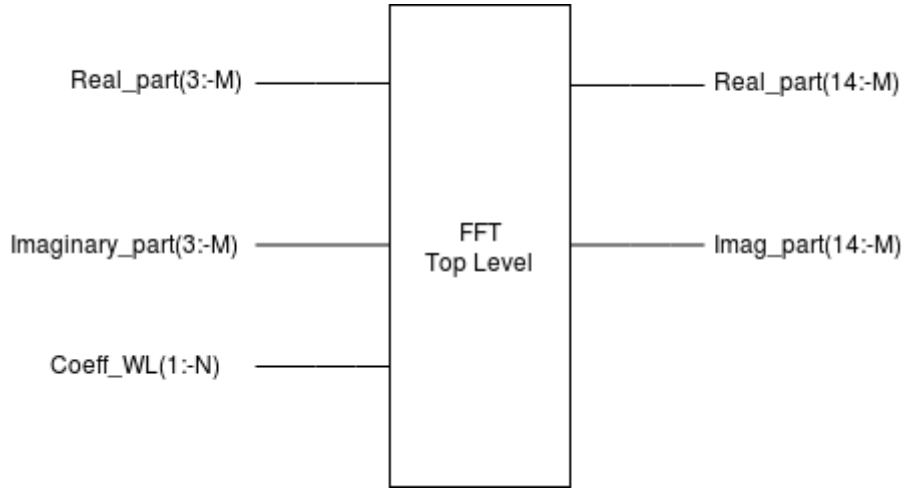


Figure (4.2): *FFT Top Block View*

Before the values are output in the final stage, the unscrambler takes the values and switches them related to the bit representation of the input values.

4.2 FFT Evaluation

In the 3rd research question described in Chapter 1 the values that were decided to be evaluated were data wordlength, coefficient wordlength and the resulting SER of the system from those 2 variables.

Wordlength is an important value especially when a system is pipelined, as it affects the cost of butterfly PEs, the chips area of implementation and the accuracy of the values.

As such an important step in the implementation is the correct determination of the wordlength and reduction of the length of the data wordlength and coefficient wordlength to get the best precision with the smallest amount of bits.

Due to cases where overflow might happen an extra bit is added in the FFT at every stage.

Data format and wordlength directly affect the accuracy of the results when the data has a floating-point or fixed point format while a longer data wordlength will increase the critical path and thus slow down the process.

Several ways to optimize the wordlength are described in [10] which consist of reducing the number of truncations and roundings, or by reducing the internal wordlength of the

FFT stages.

To calculate the system design area the computation of an FFT is split into 3 consecutive processes:

1. Reading the input values
2. Performing the FFT
3. Writing the results

Although the noise and impact of wordlength in this case is calculated directly in the 5G system, it can also be mathematically calculated by using the formula below, especially between stages if it is needed.

The round-off noise σ_B at the two outputs of a butterfly interference of an FFT output can be obtained from adding the noise sources from N_1 elements.

As such the values of the butterflies input do not need to be quantified.[4]

$$\sigma_B^2 = 2 \frac{Q^2}{12} \quad (4.2.1)$$

Where Q represents the data quantization step.

To calculate the wordlength, it is important to have the value of the noise at the output σ_e calculated:[4]

$$\sigma_e^2 = \sigma_B^2 \sum_{s=1}^M (2^{M-s}) \left(\frac{1}{2^2}\right)^{M-1-s} \quad (4.2.2)$$

where $M = \log_2 N$, and s represents the current stage.

4.3 5G Implementation characteristics

In Chapter 2 several parameters were described for the implementation of the 5G system, to have an as general as possible implementation, to this length following parameters were decided upon:

Table (4.1): 5G Implementation Values

Parameter	Value
FFT Points	2048
Waveform	CP-OFDM
Modulation	256-QAM
f_c	0.45-6 GHz and 24-52.6 GHz
Subcarrier Spacing	60 KHz
Number of active subcarriers	1156

Although the 5G system can simulate different modulation types, the official standard only uses CP-OFDM both on uplink as well as downlink, thus the decision was taken to not to test other waveforms.

In the beginning, it was also decided to use a 1024-QAM Modulation but the simulator does not support it and thus the 2nd and 3rd highest 64-QAM and 256-QAM modulation types were chosen instead.

A number of 1156 active subcarriers are used in this case with the rest being used for pilots and guard bits.

Going back to Table 2.1 it can be concluded that the choice of using a subcarrier spacing of 60 kHz would allow the system to have the most general implementation, as this allows access to both Frequency Band of 0.45-6 GHz as well as the mm-Wave band of 24-52.6 GHz.

The system will always send the same packet containing 10 symbols so that the wordlength noise can be deciphered from other values.

CHAPTER 5

SIMULATION

5.1 Introduction

The 5G simulator is based on an Open Source simulator created by Qamcom[1], which at the moment, includes waveform simulations for CP-OFDM, W-OFDM, UF-OFDM, DFT-S-OFDM, FBMC-OQAM, FBMC-QAM with hardware impairment characteristics applied to these waveforms.

The entire simulator is written in object-oriented programming in Matlab code so the next part will discuss aspects relating to it and how it works in combination with ModelSim.

The simulation will be done using co-simulation between ModelSim and Matlab and will follow the simulator block, which is a simplified block view of the OFDM block presented in Figure 2.1.

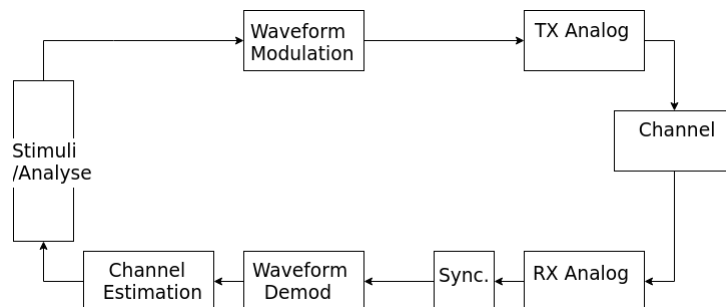


Figure (5.1): *Simulator blocks*

At the beginning when the goals were set for the thesis it was decided that from an open-source 5G Matlab simulator the Simulink model will be extracted and made to look as it is specified in version 15 of the 3GPP standard.

During the development of the Simulink block, it was concluded that following this path could not be able to be achieved, due to a combination of time constraints and problems when using the Simulink S-Function blocks and Function blocks.

So instead it was decided to have co-simulation between the Matlab and ModelSim where the input data for the IFFT/FFT blocks are taken from the workspace of the Matlab Simulation and sent to the FFT HDL block where it is applied to the values and taken back to the Matlab workspace to calculate the SER of the entire system.



Figure (5.2): *Schematic Connection between Workspace and Simulink*

To achieve this data transfer 2 scripts were written in C++ to read the data and format it for Matlab and Modelsim and back as the data could not be used directly.

5.2 Block description

The simulation blocks from Figure 5 will be described in detail starting from the stimuli block which sends the initial packet, in our case it being only one packet, which contains 10 symbols. The same packet will be sent for both modulation types, and as such it will have the same form for both 64QAM and 256QAM.

The analog blocks (RX/TX) are responsible for adding hardware impairments to the system such as oscillator phase noise, power amplifier non-linearity and carrier frequency offset while the receiver is responsible with the timing and frequency synchronization performed.

Waveforms for all received signal subject to phase noise, CFO and multi-path channel with AWGN is modeled as:[3]

$$y[n] = e^{j(\phi[n] + 2\pi\epsilon n)} \sum_{l=0}^{L-1} h_l x[n - l - v] + w[n] \quad (5.2.1)$$

where ϵ denotes the CFO, ϕ the phase noise at the receiver, x -transmission signal is the l -th tap of the channel(CIR), L -channel length, and w the AWGN.

CP-OFDM has the simplest modulation expressed as:

$$X = [F_{cp} \ F]^H s, \quad (5.2.2)$$

where s denotes the $N \times 1$ column vector consisting of the subcarrier symbols, F denotes the $N \times N$ DFT whose element is given by:

$$\frac{e^{-j2\pi kn}}{\sqrt{N}} \quad (k, n = 0, \dots, N - 1) \quad (5.2.3)$$

F_{cp} consists of the last N_{cp} columns of F (where N_{cp} denoting the CP length). CP-OFDM demodulation matrix is given by $[0, F]$, where 0 is an $N \times N_{cp}$ zero matrix for CP removal.[3]

5.3 Matlab Simulation Parameters

For our 5G system simulation, the following parameters were chosen:

Table (5.1): System simulation parameters

Parameter	Value
FFT Points	2048
Modulation	64-QAM 256-QAM
nPayloads	1
f_c	6 GHz
nPackets	1
Bandwidth	120 MHz

As described earlier in Chapter 4 the modulation will be 64 and 256-QAM instead of 1024-QAM. The number of symbols sent are 100 and the carrier frequency was chosen around 6 GHz so that it will be as general as possible and be able to function with older

systems, although it would have been interesting to test in the mm-Wave range (around 26 GHz) which is the frequency used in the European Union[11].

The MSE is calculated in the simulator for each symbol in the number of packets which is done by comparing the mean values of the equalized symbols with the number of transmitted symbols, which is defined by the following formula[1].

$$MSE = 10 * \log_{10}(\text{mean}(\text{abs}(vEqualizedSymb - vTxSymb).^2)) \quad (5.3.1)$$

While the total SER for the system is calculated by comparing the demapped symbols with the ones transmitted such that it is as close as possible to the original signal, and as such we get[1]:

$$SER = \text{sum}(vDemapSymb \neq vTxSymb) / \text{length}(vTxSymb) \quad (5.3.2)$$

Thus with the previously described values, the following values describe the system for the CP-OFDM simulation, with the original FFT block of the Matlab simulator:

Table (5.2): Simulation results 64-QAM

Modulator	64QAM
Simulated waveform	CP-OFDM
SNR	0.000000
MSE	-68.879851 dB
SER	0.000000

With the corresponding spectral regrowth of the system:

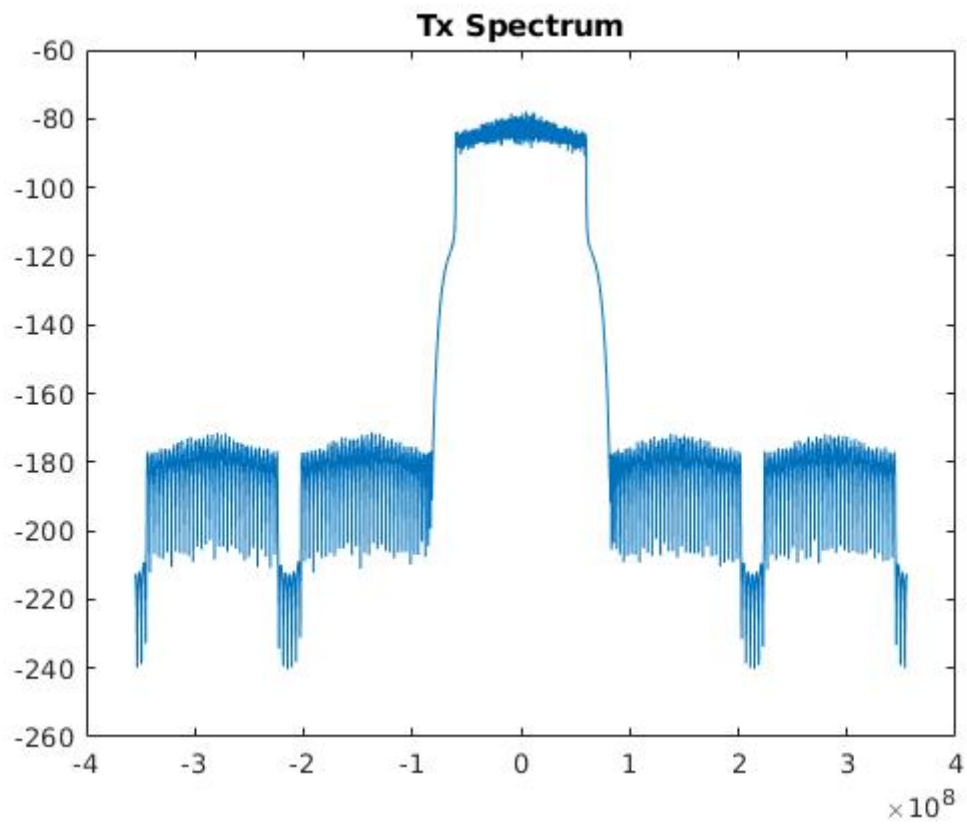


Figure (5.3): *Spectral regrowth of 64-QAM*

As well as the described payload and its modulation:

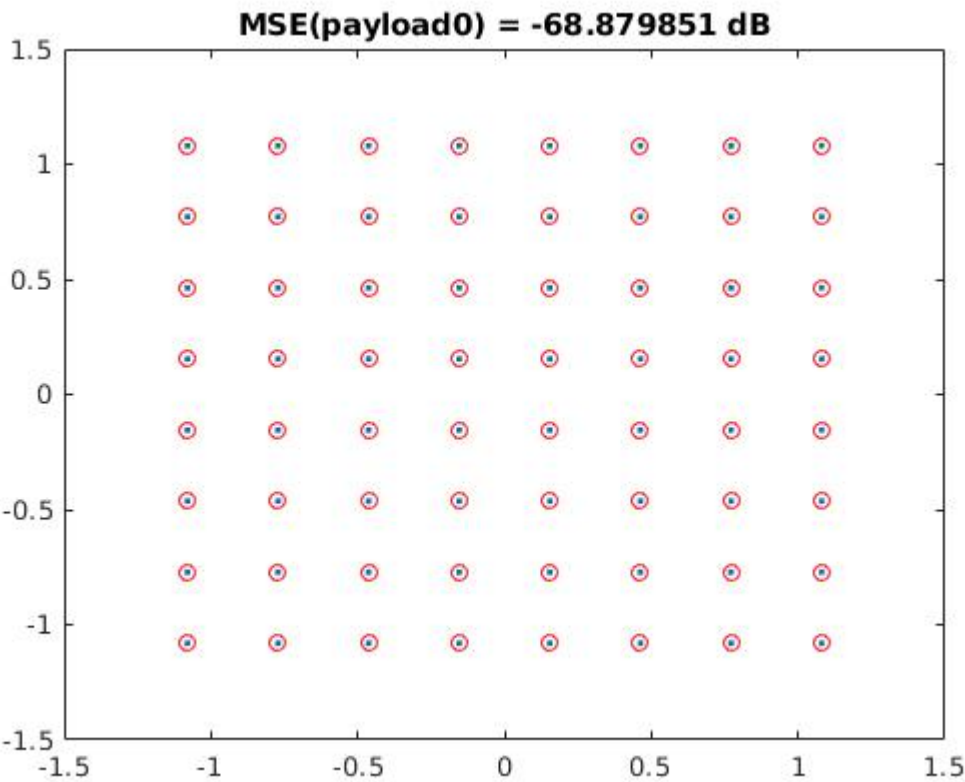


Figure (5.4): Modulation distribution of 64-QAM

And the corresponding 256-QAM modulation values:

Table (5.3): Simulation results 256-QAM

Parameter	Value
Modulator	256QAM
Simulated waveform	CP-OFDM
SNR	0.000000 dB
MSE	-77.726517
SER	0.000000

With the corresponding spectrum and modulation distribution:

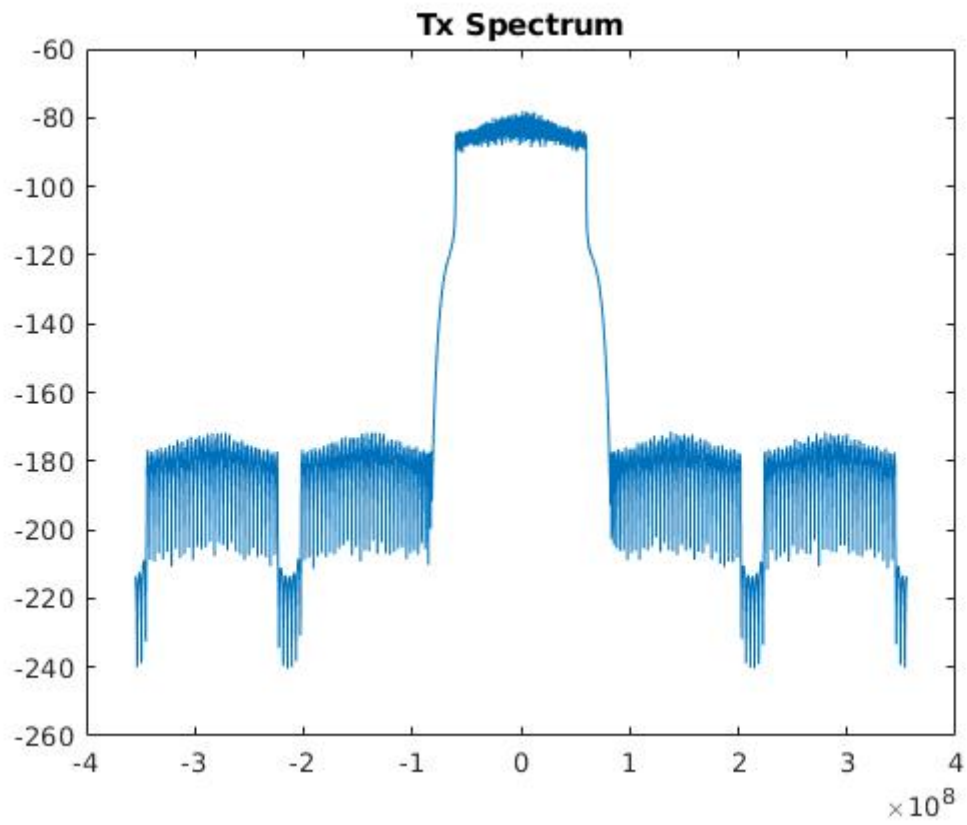


Figure (5.5): *Spectral regrowth of 256-QAM*

As well as the described payload and its modulation:

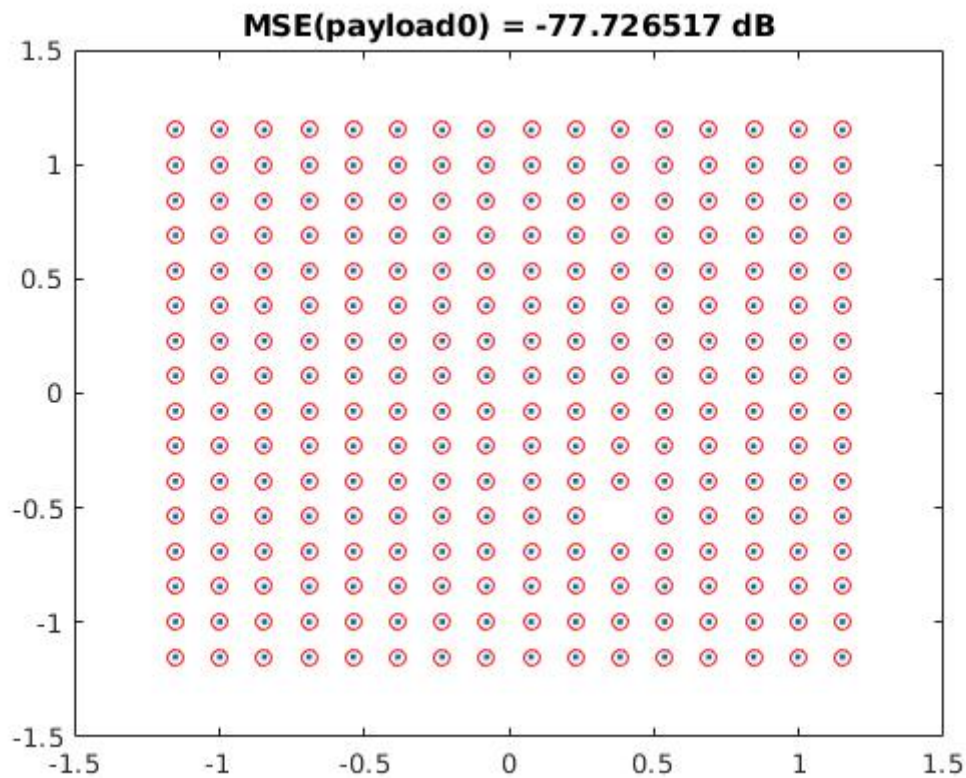


Figure (5.6): *Modulation distribution of 256-QAM*

With the simulations which describe the spectrum of the CP-OFDM and the MSE characteristics of the system.

These results will be used as benchmark values to compare to the ones that will be achieved through the system simulation with the designed FFT block.

These simulations were done with no hardware impairments and AWGN and enabling the frequency synchronization module which is performed in practice before demodulation of the receiver and will allow the system better comparisons when contrasting with the implemented FFT block.

5.4 Modelsim Results

The resulting simulations achieved through Modelsim of the VHDL FFT Implementation and the schematic of the testbench of the VHDL Code will have the following format:

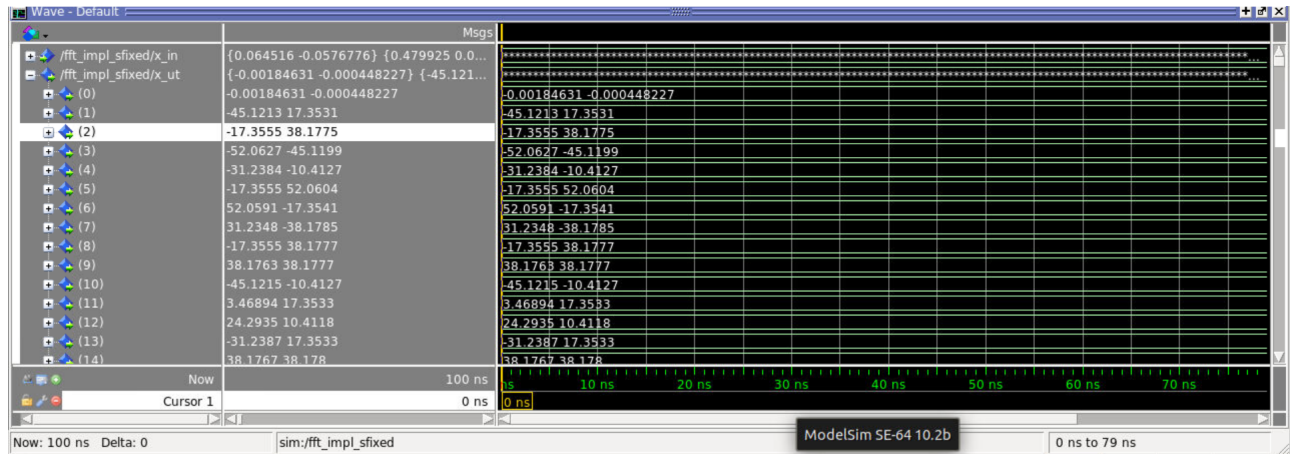


Figure (5.7): Example of a FFT simulation

Where the values are afterwards written with a testbench and transposed for use in the 5G simulator.

As described earlier in Chapter 4 in each stage of the 11 used for each FFT, the integer wordlength is increased by 1 to assure that there will not be any data overflow.

Although this was done, each stage's values were saved individually, and the actual integer wordlengths that were used by the FFT did not need. more than 8 bits, compared to the expected end length of the integer wordlength of 14.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Method

The main path in testing and verifying the accomplished system is done by comparing the benchmark values that were simulated with the ones achieved when using the VHDL FFT Block with the expectation that similar values will be achieved.

As described in Chapter 5, the parameters in Table 2.2 will be used in the realization of the simulation and the results expected are those of Table 5.

Another important aspect that needs to be discussed is the Synthesis results of the VHDL system, as well as practical aspects related to co-simulation between the VHDL-Simulink block and the original VHDL block.

6.2 Results

Initially, when discussing the wordlengths that should be tested it was decided for an interval between 10 and 24 for the wordlength and 8-24 for the coefficient wordlength.

A shortcoming of the simulation is the fact that it will only test one packet and not several as it would be in a realistic model, as in a more realistic simulation model multiple packets would be sent.

As an example the initial simulator would use 100 packets to be sent between the input and output.

The decision to reduce the number of packages was to reduce the workload due to the large amount of simulations needed to test such a system(1600 simulations for each point).

This resulted in the following graphics which display the QAM points alternation for both 64 and 256 depending on wordlength/coefficient wordlength.

The following values being representative for 64:

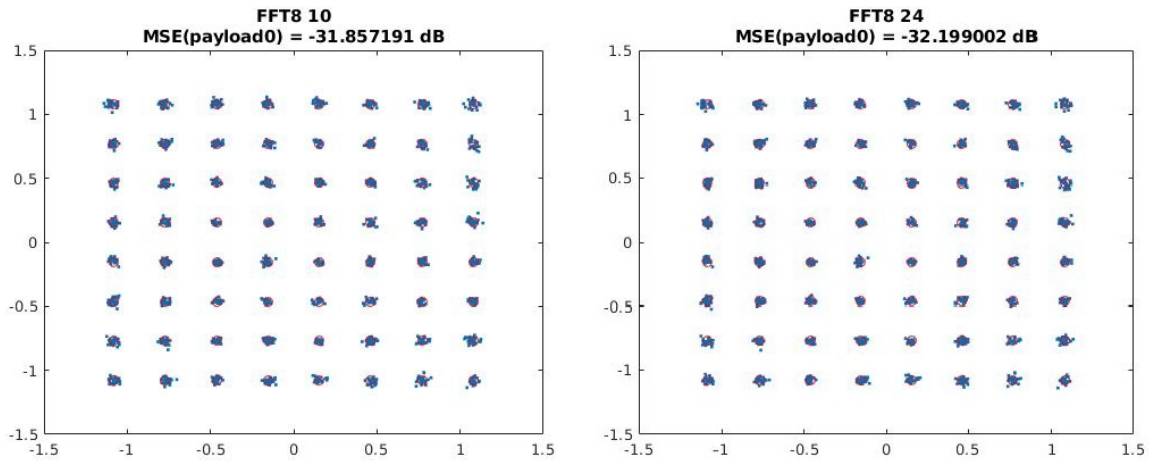


Figure (6.1): *FFT Result for WL10, WL 24, CWL 8 and 64QAM*

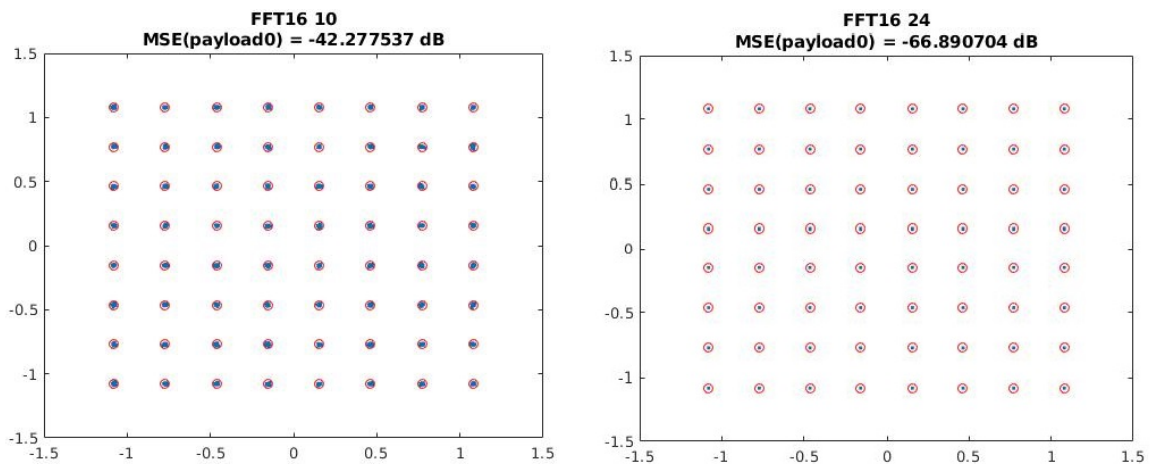


Figure (6.2): *FFT Result for WL10, WL 24, CWL 16 and 64QAM*

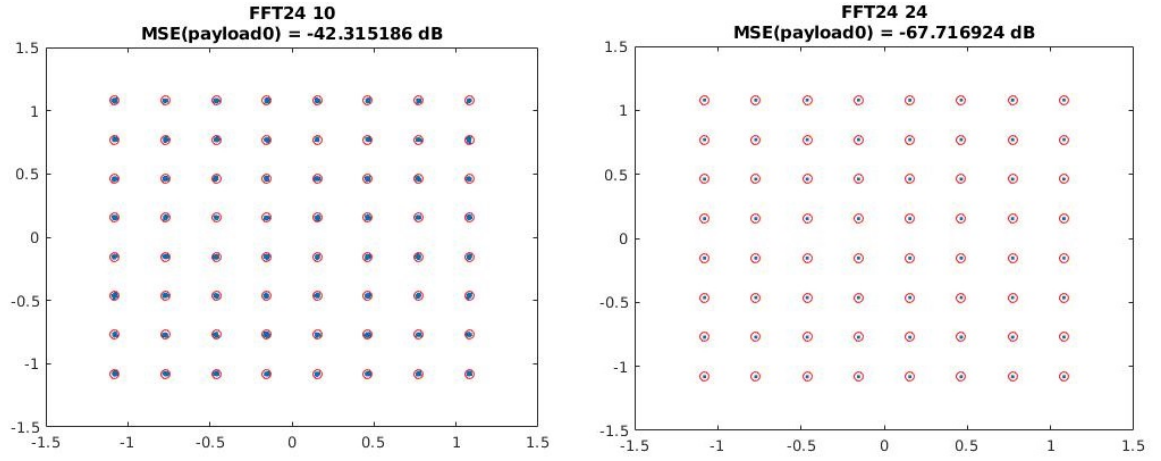


Figure (6.3): *FFT Result for WL10, WL 24, CWL 24 and 64QAM*

And further, the values which are representative for 256.

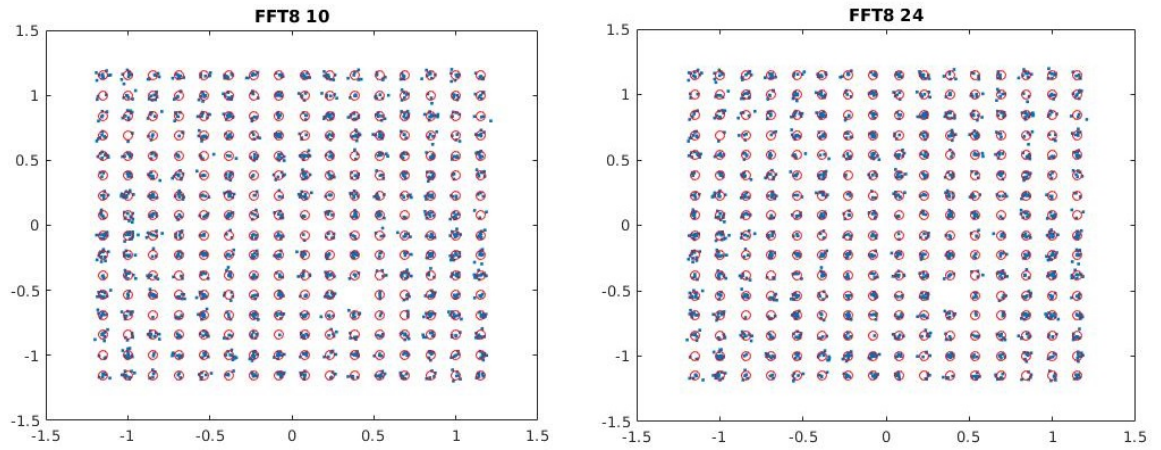


Figure (6.4): *FFT Result for WL10, WL 24, CWL 8 and 256QAM*

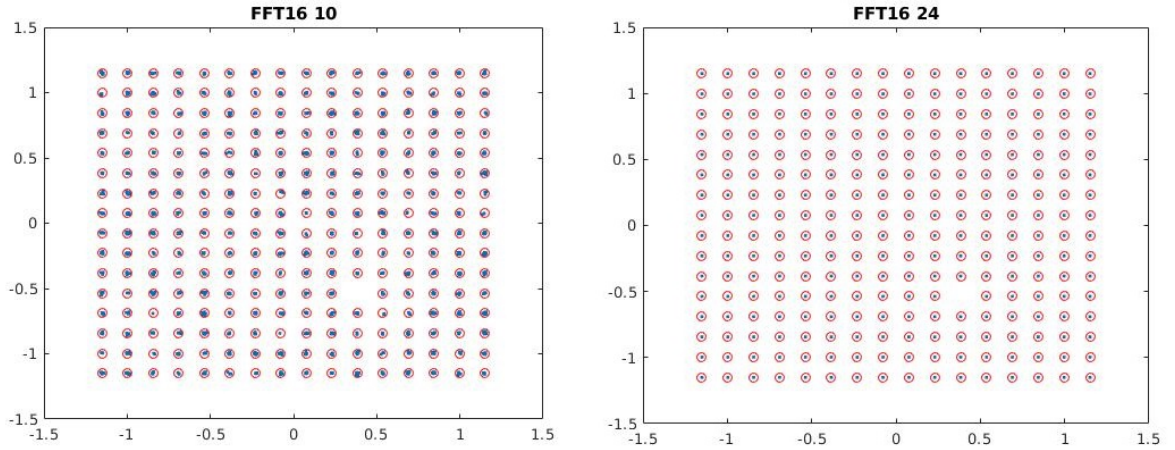


Figure (6.5): *FFT Result for WL10, WL 24, CWL 16 and 256QAM*

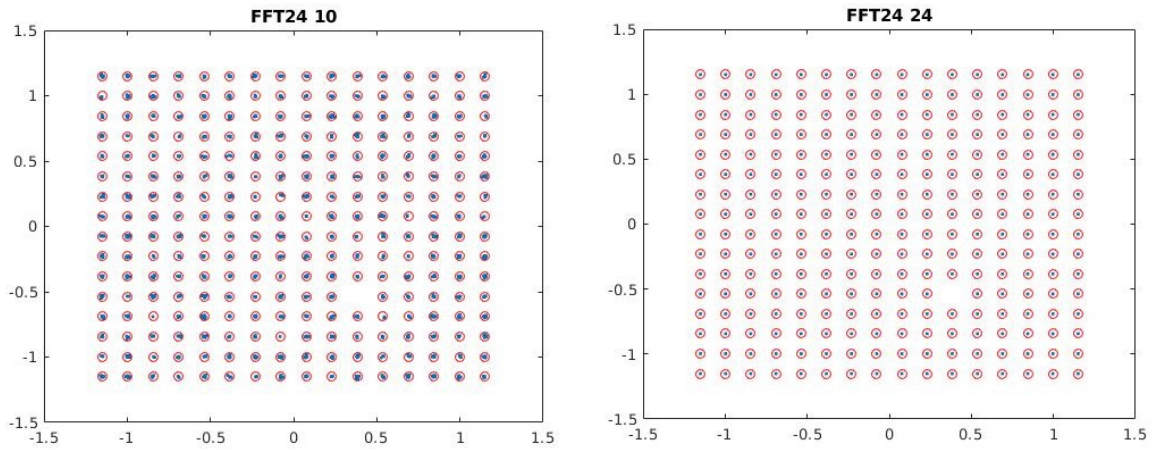


Figure (6.6): *FFT Result for WL10, WL 24, CWL 24 and 256QAM*

As can be seen, the values initially are chosen for WL and CWL assure an SER of 0, and the points are close to their expected values.

Thus the decision was taken to further decrease the assigned WL and CWL to find the point of alternation for the SER.

As such the coefficient wordlength will now alternate between 4-8 and data wordlength 6-24. The system simulation results of the system using co-simulation with the new values are described in the figure below where:

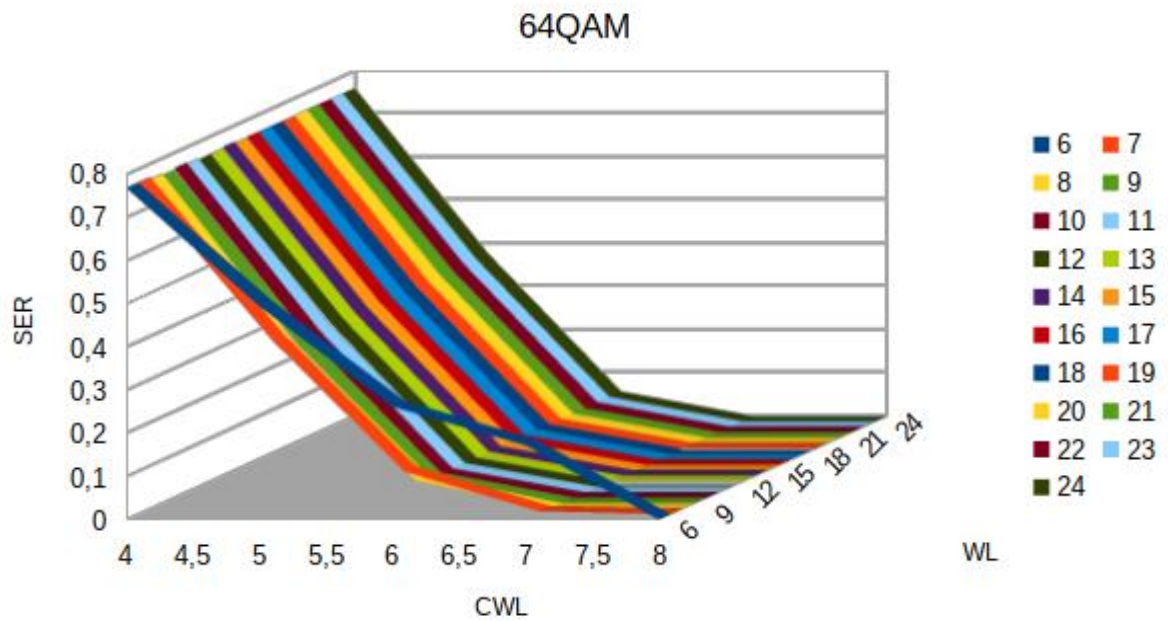


Figure (6.7): *SER values compared to CWL and WL for 64QAM*

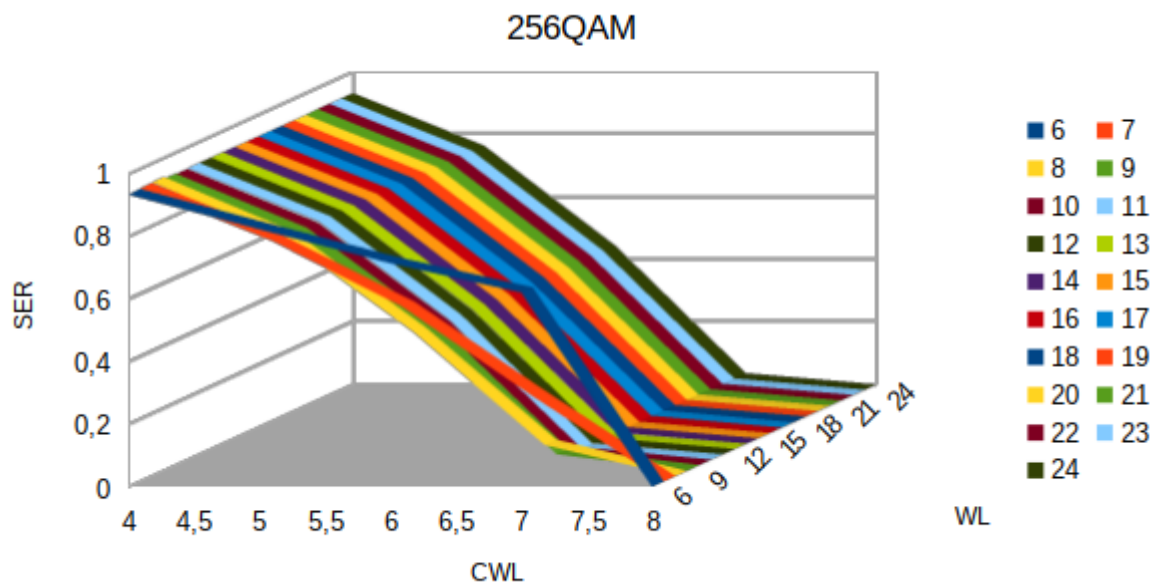


Figure (6.8): *SER values compared to CWL and WL for 256QAM*

The 2 figures describe the SER when alternating the coefficient wordlength and data wordlength. As can be seen, the lowest coefficient wordlength of 4 sends only wrong symbols, as the SER has a value of 0,769545 for 64QAM and 0,935239 for 256QAM.

64QAM		6	7	8	9	10	11 WL
	4	0,769545	0,768378	0,764294	0,760793	0,764294	0,761377
	5	0,495333	0,405484	0,371062	0,376313	0,375729	0,372229
	6	0,262544	0,0986	0,063594	0,058926	0,055426	0,056009
	7	0,172112	0,007585	0,000583	0	0	0
	8	0	0	0	0	0	0
Coeff. WL							
256 QAM		6	7	8	9	10	11 WL
	4	0,935239	0,938156	0,939323	0,932905	0,931739	0,932322
	5	0,822637	0,770712	0,762544	0,760793	0,759627	0,763127
	6	0,715286	0,55951	0,45916	0,454492	0,445158	0,447491
	7	0,621354	0,273046	0,0986	0,051342	0,04084	0,035006
	8	0	0	0	0	0	0
Coeff. WL							

Figure (6.9): SER Values for 64QAM and 256QAM

In both cases, the most important values relate to the lowest number of bits needed to achieve a correct result after the error correction is applied. As such for 64QAM it is visible that a 0,007585 SER can be achieved when having a WL of 7 and CWL of 7, while 256QAM requires a WL of 8 and CWL of 8 to receive an SER of 0.

An interesting discovery is a fact that each CWL iteration almost halves the SER and that a larger than 8 CWL will provide a high enough precision combined with a WL of 8 or larger. Another important aspect is that after WL of 16 the SER doesn't suffer any changes and as such a longer length would not be needed.

Thus designing the FFT for 8 bits WL and CWL will allow the best chip area and implementation for the used 5G system.

6.3 The work in a wider perspective

The FFT is a valuable block in the implementation of the 5G networks and will continue to be, as technology evolves and the need for bigger communication systems increases. A good implementation for FFT is not limited to 5G systems, and at the moment some examples of areas where research is done are the following:

1. Optical Communication [12]
2. Quantum Computing FFT [13]
3. FFT for large sampling rates [14]

And as such continuous work should be done in having fast implementations of the FFT which also have chip area efficiency and reduced noise from each butterfly addition.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The goal of designing and implementing a 2048-point FFT in VHDL for 5G Networks was achieved, while also answering all 3 research questions which were originally described.

Some steps are taken to simplify the design due to time limitations, mostly related to noise on the channel, as described earlier in chapter 4, while changing the original Matlab FFT block with the one designed in VHDL allowed for the same results.

The system created through the combination of a FFT block and a 5G simulator allowed also for a perfect SER with a minimum of 8 CWL and 8 WL, which was better than the initial assumption, as it was initially thought 8 CWL would be on the lower side of values having a high SER, and that the required CWL would be at least 16.

The decision could have been taken to test an even lower modulation constellation (32 QAM), but in the end it was decided that due to the distance between points as well as the data taken from the measurements for 64QAM and 256QAM, that having an even larger distance between points would only mean that an SER of 0 would be achieved faster.

7.2 Future Work

A first introspective look can be taken on the thesis as several times during the development there were a number of restrictions in its design leading to the creation of a simplified model which was chosen and as such a first step would be to pursue these restrictions and attempt to implement them by adding a channel model to the simulator.

Another interesting step could have been to alternate the number of FFT points as the system used the maximum number of points allowed by the simulator which was 2048. An interesting comparison could have been done between 1024 points and 4096, as following current trends of FFT size increase described in 2.2 we can conclude that there is a constant need for larger FFTs in communication systems.

The past year's trend shows that the lifetime of a 3GPP iteration release is around 5 years, which correlated with the increase in the bandwidth of signaling transmissions and corresponds to carrier aggregation to higher levels. Thus in the future, we can expect the need for fast FFT Implementations to increase, with the increased need for data rates.

Table 2.2 which describes the future predicted system parameters, contains a clear trend of peak data rates needed in the future and thus having faster components as the FFT should be a priority, in the development of communication systems.

The architecture could also be of interest as it was decided to use a Radix-2 to reduce the complexity of the design, but of interest could have been the use of Radix-4 or 16, which could affect the intermittent values and as such minimally require more bits to be added to some FFT-Levels.

Table (7.1): *Projected Cellular Data Rate Evolution According to Current Trends and Bandwidth Necessary to Meet These Rates.[8]*

Time	Peak data rates (in Gbps)	Bandwidth necessary (W in GHz)					
		256-QAM ($M = 8$)			1024-QAM ($M = 10$)		
		$L = 2$	$L = 4$	$L = 8$	$L = 2$	$L = 4$	$L = 8$
Dec. 2020	10.1	1.07	0.54	0.27	0.86	0.43	0.22
Dec. 2022	29.8	3.17	1.59	0.79	2.54	1.27	0.63
Dec. 2025	151.1	16.06	8.03	4.02	12.85	6.43	3.21
Dec. 2030	2259.9	240.22	120.11	60.05	192.17	96.09	48.04

The [11] standard also suggests that in the future 1024-QAM will be added, which will cause higher amounts of SER due to the increased distance between the points, and would be of great interest in case of further research.

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